

ContAQT: Designing an Interactive Data Display to Make Multi-Pollutant Air Quality Data Accessible

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Abstract

Air pollution poses critical health and environmental risks, yet public communication of air quality data is typically reduced to a single Air Quality Index (AQI) value that obscures underlying complexity. Existing visualization tools reinforce this oversimplification, leaving key AQI messages unaddressed. This paper presents ContAQT, an interactive platform that helps public audiences explore multi-pollutant air quality data in informal learning settings. Using a learner-centered design (LCD) approach, we conducted a three-step study: (1) surveys with novices and interviews with experts to identify pitfalls in current AQI communication tools, (2) iterative design and evaluation across multiple design cycles, and (3) evaluation at a local science festival. Findings show how interactive visualizations can address persistent misconceptions, spark inquiry, and support sense-making around air quality data. We contribute ContAQT as a public data display and derive design implications for building interactive systems that make complex scientific data accessible in informal learning spaces.

CCS Concepts

• **Applied computing** → **Interactive learning environments**; • **Human-centered computing** → **Information visualization**.

Keywords

Informal Learning, Interactive Data Visualization, Scientific Communication

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1 Introduction

The World Health Organization has listed air pollution as the largest environmental health risk [37]. Most public interaction with air quality data in the United States comes through the U.S. Air Quality Index (AQI), a simplified metric maintained by the U.S. Environmental Protection Agency (EPA) [3] for reporting air pollution concentrations. The AQI is presented on a color-coded 0-500-point scale and offers a quick snapshot of daily risk to humans [3, 41]. This information is typically communicated to public users through simple interactive data visualizations (see Figure 2).

While the AQI is effective for short-term alerts, its simplification hides details about the relationship between the AQI and the six pollutants used in its calculations (see Section 2.1) [14]. As a result, novices often misinterpret the AQI, leading to a fragmented understanding of what AQI really means and how pollution is measured [4, 21]. The persistence of these misconceptions problematizes the current visualization tools used for public communication (see Figure 2), which present the AQI in similarly simplified ways. To advance public understanding about air quality beyond the simplified AQI, research needs to uncover the challenges faced by novices in interpreting AQI visualizations in order to create better science communication tools.

This paper presents our design of a public-facing data display named Contextualized Air Quality Timelines (ContAQT) that presents multi-pollutant air quality index data. Using a learner-centered design (LCD) approach [50], we first conducted interviews and



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surveys of novices and experts to identify five specific tensions that exist in current tools for trying to communicate air quality data. Then, we iteratively designed three data visualization features to address these tensions by making expert ideas about AQI more transparent, while also reducing the common chart interpretation difficulties that public users have. After the iterative design, the efficacy of the ContAQT platform in resolving these tensions was evaluated in a local public science festival. In this paper, we address the following research questions:

- **RQ1:** What are the pitfalls of existing tools designed to communicate air quality data to novice users?
- **RQ2:** How can interactive data visualizations be designed to make scientific air quality data more accessible and support novices' exploration in informal public learning spaces?

By addressing these research questions, this paper makes the following contributions to designers and science communicators who want to build interactive systems to make scientific data accessible, meaning not only available but also comprehensible and usable [56]. First, we identify and synthesize an interpretive gap between novices' struggles and experts' communication goals in public-facing air quality visualizations. By examining how air quality data is intended to be communicated versus how it is actually understood, we reveal limitations of existing tools and articulate broader tensions between scientific fidelity, accessibility, and engagement in public science communication. Second, we present an interactive data visualization system and design strategies for supporting novices in exploring complex scientific data within informal learning contexts. Together, these contributions advance HCI research by clarifying what challenges matter in public data sensemaking, why they matter, and how interactive systems can be designed to support more transparent, meaningful, and accessible engagement with complex scientific data.

2 Background

The design objectives guiding our work were informed by prior work on the limitations of the single-pollutant AQI model, challenges of visualizing scientific data to the public, and efforts to make scientific data accessible in informal learning spaces.

2.1 Limitations of Single-Pollutant AQI Model

The U.S. AQI, maintained by the U.S. Environmental Protection Agency (EPA), is calculated by computing a normalized value for five major pollutants: ground-level ozone, particulate matter (at both 2.5 micron and 10 micron sizes), carbon monoxide, sulfur dioxide, and nitrogen dioxide on a 500-point color-coded scale ranging from green ("Good") to maroon ("Hazardous"), based on the concentration levels at which each pollutant becomes increasingly dangerous to humans [3, 41]. The single pollutant with the highest normalized value in a 24-hour period is the *driver pollutant*, and that number alone is reported as the U.S. AQI for that day [3].

It is important to note that single-pollutant AQI only presents the value of the highest among the six EPA-monitored criteria pollutants in a 24-hour period. This means that variation and concentrations of other criteria pollutants are obscured. As each pollutant is linked to unique at-risk populations [31], this obscurity can be meaningful. A 2017 study found that PM_{2.5}-attributable

cardiovascular health effects occurred on dates other AQI pollutants were listed as the driver-pollutant, indicating that a single-pollutant AQI alone may not communicate short-term health risks especially when pollutants such as PM_{2.5} are not listed as a driver pollutant [41]. Researchers have questioned the ability of the single-pollutant AQI to adequately articulate the health impacts of air quality [41, 58], and have also argued for disaggregation of air quality information to help the general public understand the meaning of AQI and how it can shape people's behaviors towards air quality in the long run [23, 41].

Therefore, the current single-pollutant AQI model is insufficient for effectively communicating the information people need to make informed daily decisions, which may lead to long-term health consequences. This misalignment calls for work that specifically assesses and addresses public knowledge gaps about AQI: what it represents, how it is calculated, how it should be interpreted, and applied in everyday life.

2.2 Challenges in Scientific Data Communication

Scientific communication plays a crucial role in daily life because it strongly influences people's decision-making [10]. When such communication is incomprehensible, people may struggle to make optimal choices. For example, households using an eco-feedback display with a red light often misinterpret consumption spikes from short-use appliances as the main energy drains, while overlooking continuous loads, undermining the goals of the system [52]. It is especially pronounced when the scientific concepts are presented through data visualization. A public media used four pie charts to compare the fatality rates of COVID-19, seasonal flu, SARS, and MERS, where inappropriate visual proximity downplayed COVID-19's severity greatly [15]. During the 2004 hurricane season, about half of the surveyed residents from the southwest coast of Florida chose not to evacuate because they misread the "cone of certainty" in forecast graphics, assuming their area would not be hit directly [9]. These examples highlight the need for scientific communication that is clear, effective, and unambiguous.

However, achieving this clarity is challenging. Visualization designed for science communication can fail in all aspects of color, shape, size, and orientation [33]. On the one hand, the public's ability to interpret visualizations is limited. Börner et al. [7] reported that in their study across three U.S. science museums, although most participants were strongly interested in STEM, they were only familiar with the most basic chart types like bar charts and maps, and struggled in identifying and understanding most other chart types. On the other hand, scientific communications created by experts are often not clear enough for the public. Xiong et al. [57] observed that experts often fall victim to the "curse of knowledge", or the inability to re-imagine the novice perspective, which makes it difficult for them to connect with broader audiences when communicating data patterns.

To address these challenges, our work focuses on improving the clarity and effectiveness of scientific communication in the context of multi-pollutant air quality data. By adopting the learner-centered design approach, we aim to engage the public with air quality data beyond the single AQI value. Our iterative design process ensures

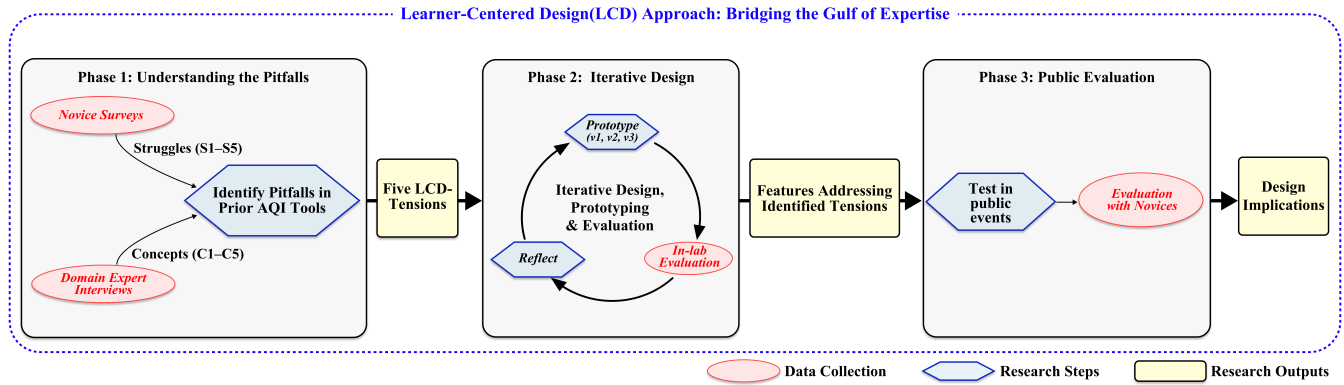


Figure 1: Overview of the paper, illustrating its three phases within the Learner-Centered Design (LCD) approach, moving from (1) understanding the pitfall to (2) iterative design and then (3) public evaluation. Red highlights indicate the research step where data was collected; blue indicates other research steps; yellow indicates the research output of each stage.

that air quality data is communicated in an accessible manner, enabling experts to convey data effectively to broader audiences.

2.3 Making Science Data Accessible in Informal Learning Spaces

Informal learning environments are settings outside of school or work where one’s task is more clearly defined, such as museums, family activities, community programs, and everyday communications [30]. In informal learning spaces, individuals learn through voluntary and interest-driven social interactions, where they often explore science interactively and engage in collaborative inquiries. Such scenarios are especially well suited to spark curiosity and encourage people of all ages, including children, youth, and adults, to engage in scientific investigation [11].

Much existing research has explored scientific data communication and engagement in informal learning spaces, with a particular focus on making scientific data “accessible” (i.e., the data can be easily interpreted) beyond “available” (i.e., the data is present and complete) [25, 56]. Block et al. designed a tree-layout visualization with the goal of reducing visual complexity while presenting the biological evolution process. The interactive data visualization was presented through a multi-user touchscreen interface where museum visitors collaboratively explore the system [6]. Ma et al. designed a visualization interface that invited users to pursue their own inquiry on time-varying microbial data [28]. Hsueh et al. proposed a tabletop exhibit with tangible cards for viewers to investigate marine animal migration through scientific encoding and comparisons [18]. Dasu et al. introduced a multi-user touch table exhibit that used narrative storytelling animation to inform the public about the microbial activity circle [12]. These concrete examples informed us on how visualization systems in informal learning spaces could engage the public to investigate data and extract scientific insights. However, there is still a lack of understanding of the clarity and effectiveness of data communication in informal learning spaces, especially on topics with complex data like air quality data.

These prior systems also highlight the broader opportunity of cultivating data literacy, which is the ability to understand and use data to inform decisions [29, 54]. This is particularly useful for domains like air quality, where we not only want the public to interpret the data but also raise awareness and take proactive, informed actions. Importantly, informal learning spaces support not only individual inquiry but also collaborative meaning-making: visitors often reason about scientific data together, compare interpretations, and negotiate explanations as part of a shared experience [13]. This social, discussion-rich context shapes how public audiences interpret complex information [47] and highlights the need for representations that support both individual exploration and group sensemaking. As multi-pollutant air quality data is a challenging and complex dataset, we want to contextually understand and address the difficulty in designing effective, accessible public data engagement while facilitating agency, collaboration, and literacy in informal learning spaces.

3 Theoretical Rationale: Leveraging Learner-Centered Design to Bridge the Gulf of Expertise

Our study is fundamentally guided by Learner-Centered Design (LCD), an approach that extends traditional Human-Computer Interaction (HCI) to explicitly foster learning and the development of expertise [50]. While early HCI focused on making computing systems “easier-to-use,” the current era has started to change the perspective of presenting opportunities for pursuing the goal to “make people smarter” [34]. This involves supporting individuals and groups to create richer and deeper understandings of content and practices, enabling them to develop expertise. Coming from this background, LCD reframes target users as *learners*, emphasizing understanding over task completion and recognizing their need for both domain knowledge development and motivation support, which distinguishes it from the classical User-Centered Design (UCD) approach [50]. A key pedagogical principle supporting LCD is “learning by doing,” where it defines knowledge to be constructed

through active engagement, artifact creation, social interaction, and reflection - all of which are especially relevant to interactive data visualization in informal learning contexts [38, 42, 51].

A central concept of LCD is understanding how to bridge the “gulf of expertise”, the conceptual gap between learners and a work domain, including the knowledge, practices, and skills learners must acquire to participate [45, 50]. In the context of data visualization, prior tools have attempted to enhance data accessibility to bridge this divide between experts and novices [14, 22, 32, 55]. However, from a learner’s perspective, these existing tools can fail to facilitate understanding and action that experts see as important. Accordingly, our study moves from diagnosing pitfalls in existing tools for novices toward a new design for interactive data visualization to better enhance environmental and data literacy for public audiences.

Building on this understanding of existing pitfalls, our LCD-driven approach first analyzes how current visualization tools succeed or fail in supporting learners’ conceptual growth. This approach connects two perspectives: (1) experts’ communication goals for AQI, and (2) learners’ struggles with current tools. By identifying the tensions at this intersection, we examine opportunities for design improvements to enhance learner understanding while aligning with expert expectations. We view our analysis of tensions as a necessary first step that can inform when and how design strategies should later be applied.

This paper consciously adopts a structure guided by the LCD approach, presenting our work as a detailed account of following our design process instead of adopting the traditional “method-analysis-result” structure. Because LCD emphasizes sequencing design around learners’ needs, our study unfolded as a sequential yet iterative process as illustrated in Figure 1. This organizational approach highlights how design decisions evolved and how each iteration advanced our understanding of supporting scientific data communication in informal learning spaces. The following sections detail this process and its outcomes.

4 Understanding the Pitfalls

To better understand the pitfalls of existing tools designed to communicate air quality data to novice users (RQ1), we conducted an investigation into novices’ experience with existing data visualization tools designed for communicating air quality data and atmosphere scientists’ priorities for communicating science concepts to the public. Our examination included a survey study with novices and semi-structured interviews with two atmospheric scientists, presented in Phase 1 of Figure 1. The study was approved by the Institutional Review Board (IRB) of the university from which the research team is based.

4.1 Method

4.1.1 Novice Survey. Data visualizations created by environmental scientists and domain experts are commonly used to communicate air quality data to the public. However, it remains unclear whether these systems effectively support novices’ understanding of AQI concepts or if audiences encounter difficulties interpreting the information they present. To inform the design of our own data visualization systems, we designed and conducted a survey study

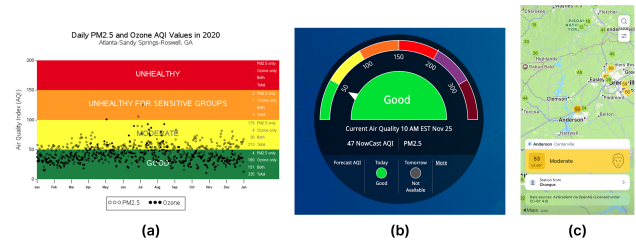


Figure 2: Existing AQI platform designs: (a) Air Data - air quality data collected at outdoor monitors across the US, www.epa.gov/outdoor-air-quality-data/air-data-aqi-plot; (b) AirNow homepage, reporting area - Atlanta, GA <https://www.airnow.gov>; (c) Map interface - screen taken from IQAir's AirVisual app. <https://www.iqair.com/air-quality-app>.

to evaluate how novices interpret AQI and the extent to which they comprehend AQI concepts and AQI-related data visualizations.

In the survey, after consent, subjects would complete questions about their past experience with air quality data, their familiarity with AQI concepts, their highest degree earned, and their age range. After that, subjects would be presented with three static data visualizations of air quality data as listed in Figure 2 that were available to the public in air quality platforms. We selected these three representations because they were created by scientific agencies or domain experts and explicitly aim to communicate AQI as an interpretable index rather than only a single number. Furthermore, while other platforms (e.g., iOS/Android weather apps) also display AQI, we chose a small, representative set that reflects distinct, commonly used visualization strategies while keeping the survey focused and manageable.

A scatter plot from epa.gov (Figure 2.a) displays AQI values in a specific location over the course of a year. The chart, as stated on epa.gov, is “useful for seeing how the number of unhealthy days can vary throughout the year for each pollutant.” The data display utilizes the AQI’s official risk category color coding to depict ranges within the graph in order to provide additional scale for the AQI values displayed on the Y axis.

The second chart (Figure 2.b) is an odometer-like radial gauge from airnow.gov displaying the current AQI in the center of a dial pointing to the corresponding risk category color range on the outer band of the dial. The textual interpretation (e.g. “Good”) is also displayed, along with a forecast for the next day and a statement of the driver pollutant.

The third representation (Figure 2.c) comes from the publicly available AirVisual mobile application and is a map-based interface with the current AQI pinned to sensor locations via color-coded markers. Users can access details about a particular location via a banner across the bottom of the screen.

After viewing each chart, subjects were asked “Do you have any questions after viewing (this chart), or is anything confusing or unexplained?” as an open-ended question.

The survey was first developed by the second author using Qualtrics [44], and a pilot study was conducted with 13 participants recruited through contacts of the research team. The research

team then revised several questions for clarity based on the pilot study feedback. We distributed the final version of the survey on social media and personal contacts of local research institutes and museums, targeting adult members of novices who were not air quality professionals in the United States. Subjects were not paid for their participation. The survey received 34 complete responses, with the average completion time of 10 minutes.

4.1.2 Domain Experts Interview. We conducted semi-structured interviews with two atmospheric science experts to identify key AQI concepts they believed the public should understand. The experts were carefully recruited for their distinguished experience in atmospheric and environmental sciences, their contributions to AQI research, and their work's impact on policymaking. The experts were recruited through the PI's professional network and were not paid for their participation.

The first expert is a principal research scientist at a major research institute at a U.S. southeastern R1 public university. They have more than 30 years of research experience on air quality, clean air policy, and social and environmental impacts. They regularly contribute to clean-air policy discussions and community-focused air-quality communication initiatives. The second expert is a professor in the Civil and Environmental Engineering Department at a U.S. southeastern R1 public university. Their research focuses on air pollution modeling, health effects of air pollutants, and air pollution control. They have served on multiple national advisory committees, including those guiding federal air-quality standards and monitoring practices, thus contributing directly to regulatory decision-making and the communication of scientific evidence to policymakers and the public.

Each interview lasted for 60–75 minutes in person and followed a semi-structured protocol aimed at identifying challenges in communicating AQI concepts to the public. We used open-ended prompts to interview the experts on (1) their work experience with air quality, (2) the pros and cons of the U.S. AQI system, (3) whether and why current U.S. AQI system is difficult for non-experts to interpret, (4) current challenges they are facing when communicating AQI concepts to public users, and (5) the core concepts they think the public should be aware of. Interviews were audio-recorded and supplemented with researcher memos.

4.2 Data Analysis

For the public survey data, the research team first excluded participants who reported both having prior experience with air quality data through their job or education and feeling confident in explaining what AQI is. These two criteria were used together to avoid over-excluding respondents: prior exposure alone does not guarantee understanding, and self-reported confidence alone can be an unreliable proxy for actual scientific comprehension. Applying this combined filter allowed us to remove participants with clear domain familiarity while retaining the broader public audience we sought to study. This resulted in 32 respondents, whom we therefore treated as novices to air quality in the subsequent analyses. Participants represented a broad age range. Among the 32 respondents, six were 18–24 years old, nine were 25–34, six were 35–44, one was 45–54, six were 55–64, and four were older than 65.

We interpret the survey results as exploratory insights into common challenges encountered by novices when using existing AQI communication tools, rather than as generalizable measures of public understanding. We then conducted an affinity mapping [27] on participants' responses to the three charts. The affinity mapping process, or the K-J Method, is a design method for generating insights and solutions through the organization of qualitative data into significant categories. There were 28 complete responses, each consisting of 3 answers reflecting participants' confusion about each chart. Among those responses, the research team first identified those containing more than one idea unit [48] and separated them into individual units. For example, the response "*What exactly is PM2.5? What do the notations on the right side of the graph mean?*" was separated into two idea units. We split 96 responses into 110 idea units. Of the 110 idea units, 48 units reported no issue of comprehension. Then, 62 effective individual units were put on separate sticky notes and sorted into groups based on similar themes. Through an iterative process, the groups were refined and consolidated in a way that some lower-level groups were compiled into higher-level themes based on their relevance to this project.

For the expert interview data, the research team inductively coded the interview transcripts using an open-coding approach to identify recurring communication challenges. The first author conducted the initial round of line-by-line coding to capture the salient ideas and considerations raised by the experts. The research team then met to review the preliminary codes and iteratively refine the code set. Through these discussions, related codes were grouped into higher-level categories, allowing the team to distill a set of emerging themes that informed subsequent important AQI concepts.

After analyzing the survey data and expert interview data separately, we synthesized these findings to investigate the pitfalls of existing tools for communicating air quality data to novices. Each of the five AQI core concepts emphasized by scientists (C1–C5) was systematically compared with the five recurring struggles reported by public audiences (S1–S5).

4.3 Findings

4.3.1 Participant Struggles with Existing Air Quality Data Representations. By analyzing the survey data, we distilled five themes from the affinity mapping process that represented the kinds of difficulties reported by participants when trying to interpret air quality data representations (see Table 1). Participants often expressed struggles with the underlying air quality terminology and scientific concepts of pollutants presented in the visualizations, underscoring the need for clearer contextual explanations on the specific domain topic. They also described confusion when interpreting visual elements such as highlights, labels, axes, and color scales, suggesting that standard data representations to experts can appear obscure to the public and require more accessible design approaches.

4.3.2 What Scientists Want the Public to Understand about AQI. By analyzing the expert interview transcript, we identified five themes that captured the AQI-related concepts discussed by the two atmospheric scientists (see Table 2). These themes summarize the perspectives shared by the two experts in our study, and the

Table 1: Novices' struggles in understanding air quality representations distilled from novices' surveys

Novices' Struggles	Explanation	Key Quotes	Count
S1: Confusion about Scientific Concepts in Air Quality.	Participants struggled with air quality terminology and scientific concepts of pollutants presented in the visualizations.	"What exactly is PM2.5 and what does Now-Cast AQI 41 mean?" (Figure 2.b)	17
S2: Struggles with Chart Topic.	Participants expressed struggles in understanding what the charts were measuring.	"Don't have any idea what is being measured" (Figure 2.c)	5
S3: Misunderstanding of Visual Encoding Choices.	Participants showed misinterpretation of visual elements such as highlights, labels, annotations, axes, marks, and the AQI color scales.	"Are the dots indicative of each day?" (Figure 2.a)	22
S4: Concerns about Data Completeness and Trustworthiness.	Participants continually raised concerns about perceived data gaps and simplification across charts.	"I'd wonder how accurate the data is" and "Not sure I need data from town to town." (Figure 2.b)	7
S5: General Usability Issues.	Participants frequently remarked on usability challenges, including readability, contrast, and layout in mobile contexts.	"Hard to read on a phone" (Figure 2.a)	11

Table 2: AQI core concepts distilled from experts' interviews

AQI Core Concepts	Explanation	Key Quotes	Count
C1: Understand AQI as a continuous scale.	AQI is not a rigid six categories but a continuous range from 0–500. The categories were decided by the AQI value.	"if you have an AQI of 99 and an AQI of 101, those aren't very different air pollution concentrations, but the messaging is extremely different." (Expert 1)	7
C2: Understand AQI as a composite value of multiple pollutants.	AQI is not one pollutant but represents six pollutants. One pollutant being the driver pollutant does not mean there are no others.	"AQI is a little bit (ambiguous) because it translates everything into a numerical value." (Expert 1)	5
C3: Be aware of the major pollutants in the living area.	Novices should be able to identify which pollutants dominate their local AQI.	"for Atlanta, the fact that they know that it's (Driver Pollutant) just being PM2.5 and Ozone, that's a good start." (Expert 2)	6
C4: Interpret AQI in local and temporal context.	While AQI tries to be representative of a general area, readings from a central city sensor may not reflect conditions in the suburbs or next door.	"understand those is that you know the air pollution happens at different scales, and how you're going to address it." (Expert 2)	8
C5: Interpret AQI impact based on different vulnerabilities.	AQI affects people differently depending on health conditions such as asthma or age.	"AQI is not a single number for everybody . . . I might be very responsive to one and others, but that might not be a problem for me." (Expert 1)	7

counts in the table represent the combined total across experts. The experts described the importance of providing contextualized explanations of AQI, emphasizing not only the simple, numeric understanding of the air quality, but a comprehensive, fine-grained perception of what AQI as a multi-pollutant index entails. The two experts also noted that, by recognizing the area's major pollutant and potential changes that would pose immediate health risks to vulnerable groups, the public could take informed, proactive decisions to protect their health and make small but meaningful changes.

4.3.3 Understanding the Pitfalls by Connecting Novices' Struggles and Experts' AQI Concepts. To investigate the pitfalls of existing tools for communicating air quality data to novices, we juxtaposed the AQI core concepts identified in the expert interviews C1–C5 and the comprehension challenges reported by novice participants S1–S5. This analysis surfaces a set of Learner-Centered Design (LCD) tensions that the tools we studied did not address well: mismatches between what experts aim to convey and what novices are actually able to interpret without domain background or data expertise.

For instance, confusion on scientific concepts in air quality S1, coupled with misunderstandings of visual encoding choices S3, created significant barriers to achieving the expert' emphasis on

Table 3: Concept-Struggle matrix with the LCD tensions that existing tools mis-emphasize

LCD Tensions	Description	AQI core concepts (C)	Novices' Struggles (S)				
			S1	S2	S3	S4	S5
Tension 1: Continuous vs. Categorical Thinking.	Experts want the public to see AQI as a continuous scale (C1); Novices often lack conceptual clarity (S1) and misinterpret visual encodings (S3), perceiving AQI as rigid categories.	C1	✓		✓		✓
Tension 2: Composite Index vs. Single-Pollutant Assumption.	Experts want audiences to see AQI as a composite index of six pollutants with one driver pollutant (C2); Novices misunderstand pollutant concepts (S1) and breakdowns (S3) as single-pollutant readings.	C2	✓		✓		✓
Tension 3: Driver Pollutant vs. Information Complexity.	Experts want people to identify common driver pollutants (C3); Novices struggle to contextualize temporal charts (S2) and decipher visual encodings (S3), obscuring how pollutants dominate over time.	C3		✓	✓		✓
Tension 4: Contextualize vs. Generalize.	Experts want AQI interpreted in temporal and local contexts, considering sensor coverage and variability (C4); Novices question data reliability and completeness (S4) and prefer an immediate, generalized report.	C4				✓	✓
Tension 5: Unclear Relevance vs. Personalized Health.	Experts want AQI to highlight personal health risks (C5); novices lack knowledge of vulnerability (S1) and distrust data completeness (S4), unsure of how it's relevant to their lives.	C5	✓			✓	✓

understanding AQI as a continuous scale **C1**. Note that **S5** was considered in all core concepts delivery because general usability issues would hinder users' smooth interaction with the interface, leading to pitfalls.

Based on these findings, Table 3 presents a synthesis of the mapping results. The table aligns expert-discussed AQI concepts (**C1-C5**) with the novice-reported struggles (**S1-S5**) and highlights the resulting LCD tensions. These tensions illustrate the specific pitfalls of existing visualization approaches. By reframing the problem as unresolved LCD tensions, we clarify how these participant-reported challenges point to limitations of existing visualization approaches and motivate the design choices behind the ContAQT platform.

5 Design of the ContAQT Platform

Addressing the design tensions in air quality data tools as outlined in Section 4, we proposed a public data display called Contextualized Air Quality Timelines (ContAQT). For this paper, we focused on the first four LCD Tensions, as Tension 5 would likely require AI-driven personalized suggestions, which are beyond the scope of this work. We first identified three features to address these tensions and then iteratively designed and evaluated these features. This section highlights the major design features of ContAQT and how they evolved through multiple iterations, as presented in Phase 2 of Figure 1.

5.1 The ContAQT Platform

5.1.1 The Core Features. The ContAQT platform was designed to be a kiosk experience in local informal spaces, such as museums, public science education events, and a nearby mixed-use area adjacent to the research institute, which features academic and commercial buildings and attracts a broad public audience, primarily adults from diverse professional backgrounds. It was intended for deployment on a 65-inch 4K touch-screen kiosk (as shown in Figure 3) to support full-scale interaction. The goal of the system was to support novice members of the general public with diverse educational backgrounds and limited prior knowledge about air quality, rather than designing specifically for children as a primary user group. Accordingly, the iterative in-lab evaluations in this section focused on novice adults, who could take part in verbal think-aloud and interview protocols.

We intentionally designed it as a free-choice learning environment, where visitors leverage their own agency to pursue questions of interest rather than receive a pre-curated story. This choice aligns with the goals of informal science learning, prioritizing active exploration over passive story consumption and allowing users to construct their own interpretations of AQI data. Accordingly, the present study in this paper centers on how visualization representations support inquiry-oriented, user-driven engagement in these settings.

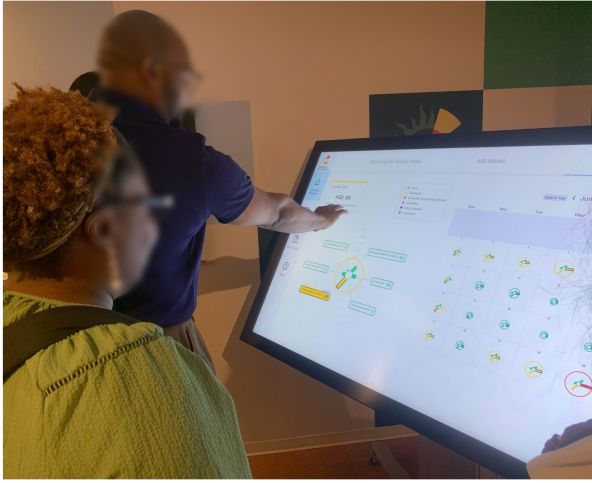


Figure 3: In-lab evaluation setup for the ContAQT system. The figure shows participants standing in front of the kiosk interacting with interactive elements on the screen during a full-scale usability session.

To address the identified LCD tensions, we designed three main features for the ContAQT platform.

- **AQI Core Chart:** a daily snapshot of all six pollutants to show how AQI is composed, addressing Tension 1 (Continuous vs. Categorical Thinking) and Tension 2 (Composite Index vs. Single-Pollutant Assumption).
- **AQI Timeline:** a time-series view that shows AQI values and driver pollutants, helping novices interpret trends and contextualize changes, addressing Tension 3 (Driver Pollutant vs. Information Complexity) and Tension 4 (Contextualize vs. Generalize).
- **AQI Calendar:** a calendar-layout monthly view that allows exploration by specific dates and comparison across days, addressing Tension 3 (Driver Pollutant vs. Information Complexity) and Tension 4 (Contextualize vs. Generalize).

The dataset used in the ContAQT platform includes daily Air Quality data of the six major pollutants collected by a government-owned air quality monitor site in a major city in the southeastern U.S. from 2014 to 2023. The dataset was constructed based on the metadata downloaded from the United States Environmental Protection Agency (EPA) Air Quality System outdoor air quality data repository.

5.1.2 The Design and Evaluation Process. We followed an iterative design and evaluation approach as described in [12]. The system consisted of three main features: the AQI Core Chart and AQI Timeline underwent three design iterations, while the AQI Calendar underwent two. After each version was implemented, we recruited new participants to evaluate all features available in that iteration to avoid carryover or ordering effects. Five participants tested the initial versions of all three features, six new participants evaluated the second versions of the AQI Core Chart and AQI Timeline, and

two other participants evaluated the final versions of the AQI Core Chart and AQI Timeline together with the updated second version of the AQI Calendar. No individual took part in more than one round.

The study was IRB-approved. Each in-lab user study followed the same protocol: participants interacted with the prototype while thinking aloud and then completed a brief semi-structured interview about their interpretations and points of confusion. All sessions were screen-recorded and audio-recorded for analysis. Participants were undergrad- or grad-students at the researchers' institution without prior experience with air quality, recruited through lab connections and word of mouth, not affiliated with the PI's lab, and received no compensation. Participants were indexed by version and ID (e.g., V2-P4 referred to Participant 4 in the Version 2 evaluation). After each study, we transcribed the audio and conducted inductive open coding on the combined think-aloud and interview data to identify recurring interpretation challenges that informed subsequent design refinements.

5.2 AQI Core Chart (Figure 4)

The AQI Core Chart was designed to address Tension 1 (Continuous vs. Categorical Thinking), and Tension 2 (Composite Index vs. Single-Pollutant Assumption) (see Table 3). This view represents air quality data for a specific day by plotting all six EPA-monitored pollutants on axes around a central point. Values are encoded through relative lengths, allowing the shape to convey both overall air quality and the structural role of the driver pollutant.

Version 1 (Figure 4.a) explored a spider-chart design that connected pollutant values into a polygon. The shape was color-coded by AQI risk category. The longest spike indicated the driver pollutant. This design aimed to address Tension 1 by showing that even when two days share the same AQI category color, the size of their shapes differs, reinforcing AQI as a continuous value. It also attempted to make explicit the link between the highest pollutant and the overall AQI, while signaling the contribution of multiple pollutants to the daily air quality, addressing Tension 2. While this version validated the importance of showing all six pollutants together, participants found the irregular polygon and shared color encoding difficult to interpret, limiting its ability to support Tension 1 and Tension 2.

To address these challenges, Version 2 (Figure 4.b) adopted a Rosa chart of six radial wedges, where wedge length encoded each pollutant's AQI, color indicated health categories, and a shaded ring marked the driver pollutant. This compact form made the composite nature of AQI more apparent, partially addressing LCD Tension 2. Yet the mixture of colors around the center left the overall AQI category ambiguous. As one participant noted, *"It's hard to understand if AQI 134 is good or bad because there is a mix of colors"* (V2-P2). This indicated that while the design presented multiple pollutants, issues remained in addressing LCD Tension 1.

Across these iterations, the design process specifically surfaced novices' persistent confusion around mixed-color encodings, uncertainty about how to read composite shapes, and difficulty relating individual pollutant values to the overall AQI. These insights directly shaped the final design. Version 3 (Figure 4.c) incorporated the lessons from Version 1 and Version 2. We simplified the Rosa

AQI Core Chart Iterations

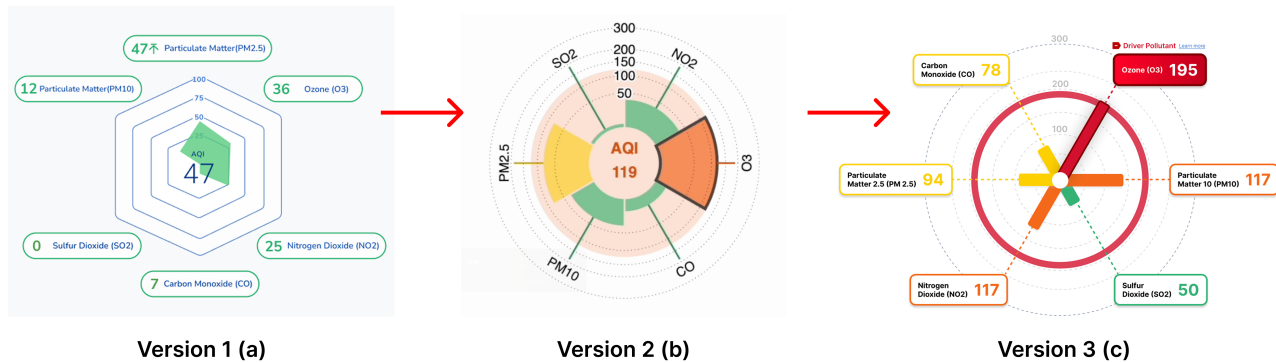


Figure 4: Iteratively designed three AQI Core Charts: (a) Version 1 uses a spider chart connecting pollutant values into a polygon. (b) Version 2 presents a Rosa chart with colored radial wedges indicating pollutant levels around a central AQI value. (c) Version 3 combines Rosa chart with bar charts to highlight the pollutant with the highest AQI contributions.

chart into consistent-width radial bars, added numeric labels for transparency, and explicitly highlighted the driver pollutant. The centered AQI marker was replaced with a circular reference ring, allowing users to compare overall AQI against individual pollutant values. These changes are directly aimed at resolving the confusion regarding the “mix of colors” and to better address LCD Tension 1 and LCD Tension 2, proving the improved effectiveness in communicating AQI values and driver pollutants.

5.3 AQI Timeline (Figure 5)

The AQI Timeline view was designed to address Tension 3 (Driver Pollutant vs. Information Complexity) and Tension 4 (Contextualize vs. Generalize) (see Table 3). It presents time-series air quality data, showing both overall AQI values and the six pollutant values, with the goal of communicating how AQI changes over time and which pollutants most often act as the driver.

In Version 1 (Figure 5.a), the Timeline view presented a daily time-series view using bars to show pollutant values (height) and risk categories (color), complemented by a line graph for daily AQI and the maximum pollutant. Buttons allowed users to switch between pollutants for comparison. This design was intended to help novices recognize driver pollutants (Tension 3) and contextualize AQI over time (Tension 4). Evaluation showed mixed outcomes. Three of five participants connected the AQI to the line chart, but most were confused about the driver pollutant, noting, for example, that “the overall AQI confused me because I thought it was the highest value of that particular pollutant” (V1-P3). This indicated that the design did not sufficiently resolve Tension 3 or Tension 4.

To reduce the confusion in Version 1, Version 2 (Figure 5.b) directly integrated the AQI Core Chart (Version 2) into the time series. Each day appeared as a bar showing pollutant level with an overlaid Rosa chart, making daily pollutant profiles visible while preserving AQI trends (addressing Tensions 3 and 4). While participants clearly

recognized AQI trends, the AQI Core Chart (Version 2) drew little attention, and several found the view information overloaded (e.g., “too much information ... have different things going on” from V2-P2). The design thus improved AQI value trend recognition but failed to clarify driver pollutants trend, motivating further changes in Version 3.

Version 3 (Figure 5.c) simplified the view to show only how AQI changes within and across ten years, limiting the view’s goal to only address LCD Tension 4. Daily AQI values were grouped by year and color-coded by risk categories on a heatmap, with each column representing one week and each row a day of the week. This layout concentrated long-term trends while enabling week-day-weekend comparisons. Participants correctly interpreted the chart as showing multi-year AQI trends and could compare across years, but one noted a desire for driver pollutant details (V3-P1). Building on the earlier iterations, this final design reflects what we learned from Versions 1 and 2: novices struggled when asked to toggle among pollutants (V1), and embedding full pollutant breakdowns for every day overwhelmed users (V2). By constraining the scope to a heatmap view, Version 3 reduces cognitive load while still supporting meaningful temporal comparisons. Together, these refinements show how the iterative process clarified the balance needed between detail and simplicity in timeline-based representations for novice audiences.

5.4 AQI Calendar (Figure 6)

The AQI Calendar feature is an “at-a-glance” monthly view of historical air quality data, designed to address Tension 3 (Driver Pollutant vs. Information Complexity) and Tension 4 (Contextualize vs. Generalize) (see Table 3). Unlike the AQI Timeline, it places small multiples of the AQI Core Chart on a calendar layout, enabling novices to explore daily breakdowns while reading patterns across a month.

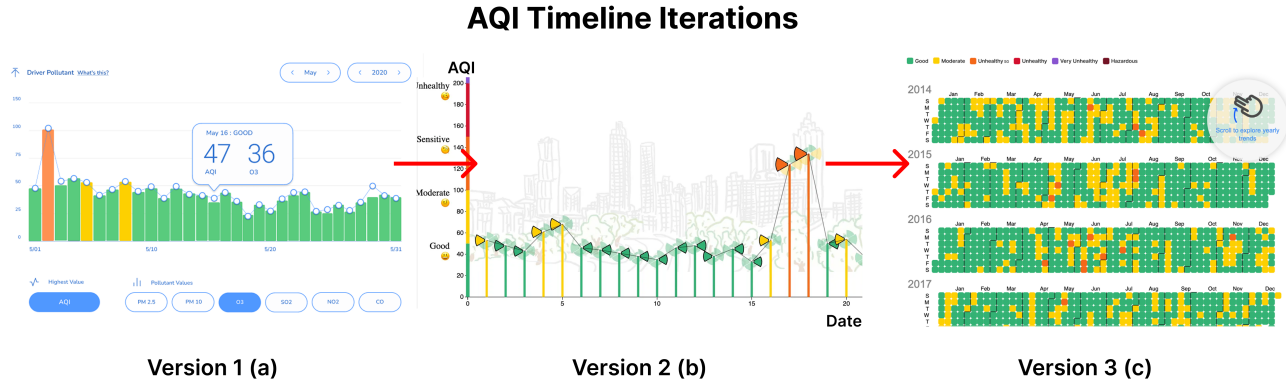


Figure 5: Iteratively designed three AQI Timeline views: (a) Version 1 uses bar and line charts to show pollutant values and daily AQI. (b) Version 2 overlays daily Rosa charts on pollutant bars to reveal pollutant profiles over time. (c) Version 3 presents a multi-year AQI heatmap for long-term temporal comparison.

Version 1 (Figure 6.a) introduced the AQI Core Chart (Version 1) into each calendar cell. While four of five participants recognized how the small glyphs corresponded to the pollutant layout in the daily breakdown, the small glyph size made it difficult to compare differences across days, and participants struggled to determine both AQI levels and driver pollutants within the dense layout. As one participant noted, “*The shapes in the calendar look like they are all saying different things.*” (V1-P5) Although the strategy itself of representing AQI core charts in a calendar was accepted by users, the current design did not address LCD Tension 3 nor LCD Tension 4, hindering the goal of making data accessible.

Version 2 (Figure 6.b) introduced the updated AQI Core Chart in a calendar layout. To improve clarity, Version 2 integrated calendar access into the timeline: tapping a date opened a view with that day’s Rosa chart alongside the monthly calendar. Participants immediately engaged with this feature, noting it was “*interesting to see that every day the circle size is changing as well as the highlighted pollutants.*” This version highlighted the importance of distributing cognitive load across views, allowing novices to browse broadly but inspect deeply when desired. Overall, these iterations show how the Calendar’s design evolved toward a balance of overview and detail to provide clearer comparative cues across many small multiples and construct layered interaction pathways.

6 Public Evaluation

The three features designed in Section 5 were synthesized into one system. We showcased the ContAQT platform through in-person events during a local science festival. The goal of this section is to understand how data visualizations can be designed to communicate scientific data to novices (RQ2). We collected participants’ comments from the events and conducted a qualitative analysis to gain a deeper understanding of their experiences. This part is demonstrated in Phase 3 of Figure 1. The study was also IRB-approved.

6.1 Setup

The ContAQT platform was exhibited at two public events during the local science festival in 2025, a city-wide celebration of over 140 activities. The first evaluation (Figure 7.a) was at the festival’s kickoff on the research team’s campus, attracting 5,000 visitors. The second (Figure 7.b) was showcased within the PI’s lab booth alongside other interactive science exhibits at a local park with 25,000 visitors. Both events are open to the public. Due to mobility constraints and venue regulations at the public events, we deployed a portable 32-inch 2K touch screen instead of the full 65-inch kiosk planned for permanent installations. The interface layout and font sizes were proportionally adjusted to ensure that all visual elements remained readable and the interaction flow remained consistent with the intended large-scale design.

6.2 Data Collection

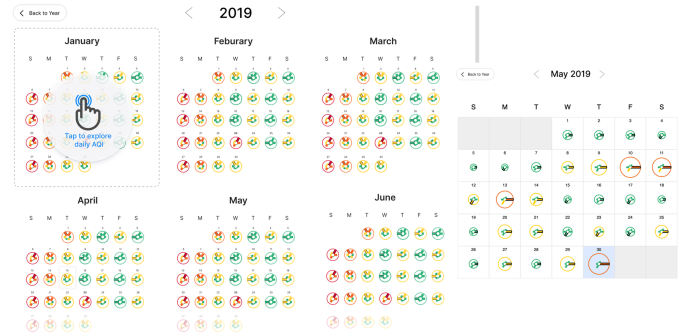
Across the two public exhibition events, the touch screen was interacted with by a broad cross-section of visitors, including young and middle-aged adult visitors and families with children. For the first event, during the four-hour exhibition, approximately 80 users interacted with the touch screen. For the second event, during the six-hour exhibition, approximately 100 participants interacted with the ContAQT platform. Through onsite observations, visitors typically spent around 3-5 minutes exploring the interactive system before moving on.

Considering the large volume of participants and the informal setting, instead of conducting interviews with participants, we chose to ask participants who orally consented to participate to leave a one-sentence comment about what questions they still had or what suggestions they wanted to give to our platform, akin to a data collection method commonly used in museums and other informal learning spaces to gather user reflections [49, 53]. For participants younger than 18 years old, we asked both the participants and their guardians for their oral consents/assents. The research

AQI Calendar Iterations



Version 1 (a)



Version 2 (b)

Figure 6: Iteratively designed two AQI Calendar views: (a) Version 1 places small AQI Core Chart glyphs in each calendar cell to show daily pollutant breakdowns. (b) Version 2 integrates an updated Rosa chart with the monthly calendar, enabling broad monthly scanning and detailed daily inspection through layered interaction.

team took field notes during both events to record open comments contributed verbally.

In total, 59 comments were received from the two events. Our goal was to evaluate how our interactive data visualization platform, ContAQT, made air quality data more accessible and supported novices' exploration (RQ2). To analyze these reflections, we applied the five themes (S1–S5) identified in Section 4 and deductively categorized the comments accordingly to assess how the system supported novice interpretation. Comments that did not align with the existing categories were inductively coded to generate additional themes, following an approach consistent with thematic analysis [8]. Specifically, while we recognize that a substantial portion of the comments came from younger participants who interacted with the touch screen during the public events, we analyzed all comments collectively as reflections from novice users, rather than excluding children's input or separating it from adults', as the goal of the public evaluation was to understand how a broad novice audience, spanning various age groups, engages with air-quality visualizations in informal learning settings.

6.3 Results

In this section, we report participants' comments for each category. Specifically, no comments were classified under S2; therefore, we did not include this struggle in the report.

6.3.1 S1: Confusion about Scientific Concepts in Air Quality. 18 comments were classified under this category. In general, comments coded here suggest that the ContAQT platform made the AQI computation process more accessible. For example, one participant noted, "it was surprising to know that AQI is not taking the average", clarifying a common misunderstanding about AQI concepts. The system also prompted participants to inquire beyond the information provided, asking "What's this unit called?" and "Are the scales of transforming from raw data to AQI the same



Figure 7: The ContAQT platform was showcased at two events during a local science festival, where visitors interacted with the touchscreens: (a) on-campus kickoff and (b) park-based booth. Feedback was collected after users interacted with the 32-inch touchscreen.

for each pollutant?", which reflects not only uncertainty but also an emerging interest in the mechanisms behind AQI computation. This pattern echoes observations from later in-lab evaluations in Section 5, where users increasingly moved from basic terminology confusion toward probing "how" and "why" AQI works. Others connected the information to broader contexts, asking "Who is included in sensitive groups?" and "Where does the pollen belong to?" Such responses indicate that the system encouraged participants to situate AQI within a frame. Rather than confusion, these responses indicate that the system enhanced curiosity and supported novices in situating AQI within broader contexts.

6.3.2 S3: Misunderstanding of Visual Channels. Four comments were classified under S3. Participants did not express confusion about interpreting the visual design itself, but instead questioned the design choices. For instance, one asked "why design in the circular (Rosa chart) way?", while two participants mentioned that they

also wanted to see a line chart showing the AQI trend. Another suggestion given was to “*add numbers of counts of types of days,*” highlighting an interest in more explicit quantitative representations. Such comments show that the system prompted participants to critically evaluate visualization design and compare alternatives that might better support their investigations. This pattern differs from our earlier in-lab evaluations, where participants often reported difficulty interpreting the Rosa chart’s encoding. The touch screen setting, with its brief, self-directed interactions, appeared to elicit more high-level design critiques than basic comprehension challenges. This shift aligns with our iterative refinements, which reduced initial interpretive barriers and left participants free to focus on the comparative merits of different chart types. Taken together, these findings suggest that making scientific data accessible requires not only simplifying interpretive demands but also supporting users’ tendencies to draw on familiar visualization conventions when exploring data in informal public settings.

6.3.3 S4: Concerns about Data Completeness and Trustworthiness. Eight comments were classified under **S4**. Participants raised questions about the availability, reliability, and coverage of air quality data, such as “*Why is there missing data?*” and “*Why is there only (city name) data?*” Others inquired about infrastructure, asking “*Where is the AQI sensor located?*” and “*How many sensors are in (city name)?*” or reflected on cross-platform consistency, “*Are AQI numbers reported by different platforms the same or different?*” These responses resonate with the concept of understanding AQI in temporal and local contexts. Rather than indicating simple confusion, these comments suggest participants’ attempts to consider data provenance and decision-making in air quality monitoring, showing how interactive visualizations can enhance accessibility and support novices’ exploration. In this informal learning setting, users engaged briefly yet still gravitated toward evaluating the credibility of the underlying data, which is an aspect that experts had also emphasized as central to contextual AQI understanding.

6.3.4 S5: General Usability Issues. Seven comments were classified under **S5**. Several participants noted that placing the two displays side by side was helpful, as it allowed them to cross-reference information across screens. At the same time, some usability issues emerged. A number of participants reported uncertainty about whether the experience had ended (“*wasn’t sure it was ended at the last screen*”) or loss of interest after reading through lengthy pollutant cards. It echoed the research team’s observation that only a few users interacted with all the information pop-up cards that contained explanations of AQI terminologies. These findings suggest that, although the content was more accessible, the pacing and information density still placed a cognitive load on users during short, self-directed interactions. In the public setting, users selectively skimmed or skipped text-heavy elements, revealing how informal environments amplify the need for immediate clarity and low-effort interaction. This means that usability in informal learning environments depends not only on interface clarity but also on offering lightweight, discoverable interaction pathways that minimize effort while still supporting deeper inquiry when desired.

6.3.5 Curiosity about data patterns. We also identified two new themes from visitors’ comments, which extend beyond the struggle-focused categories S1–S5 and capture how visitors used the ContAQT system with the touch screen setup to explore patterns in the data and situate AQI within broader societal contexts. One new theme that emerged captured participants’ curiosity about the patterns they identified through their usage of the system. 19 comments fell into this category. Rather than focusing on air quality terminologies or visual encodings, participants attempt to interpret data patterns observed from the visualizations and the reasons behind them. They asked questions such as “*Is there any seasonal trend in AQI?*” and “*Is there any AQI difference during day and night?*” Others looked beyond the data provided, asking “*How is the AQI like in other cities?*” or “*Is (city name)’s overall AQI good in other years as well?*” Some suggested predictive or comparative features, such as “*adding forecast*” and “*find the worst days and compare across.*” These responses show that ContAQT encouraged novices to search for temporal, spatial, and comparative patterns, demonstrating how interactive visualizations can make data accessible and spark exploration. Notably, this pattern-seeking behavior extends beyond the struggle-oriented themes (S1–S5) and reflects a more exploratory form of engagement, where visitors moved from interpreting individual visual elements to reasoning about broader trends and comparisons in the data.

6.3.6 Framing AQI in societal contexts. Another theme reflected participants’ efforts to consider the air quality issue from a broader societal scope rather than on an individual scale. Several comments suggested connecting air quality data with collective actions to improve the air quality, such as “*how can we take action*” and “*have an action call for each health degree*”. Others emphasized broader contexts, linking air quality to “*wildfire*”, “*hospital visits and health outcomes*”, or “*policy differences across cities.*” Some also suggested integrating the system into weather apps. These responses show that ContAQT prompted audiences to situate air quality within health, policy, and collective decision-making, illustrating how interactive visualizations can extend accessibility and exploration to broader societal contexts. Such societal-level reasoning also goes beyond the novice struggles identified earlier, indicating that visitors were not only interpreting AQI but actively contextualizing it within broader environmental and policy conversations, which is also an interpretive layer that aligned with expert-emphasized AQI concepts but rarely surfaced in earlier design phases.

The emergence of these new themes also suggests that, compared with the previous visualizations for air quality data for public communication we surveyed, the ContAQT platform inspired participants to engage in deeper and more expansive reflections about air quality data, not only interpreting what was shown on the screen but also reasoning about why certain temporal or spatial patterns might occur. Visitors moved beyond surface charts or terminology comprehension toward more open-ended inquiry in collaboration with other visitors, and connecting AQI to broader societal, environmental, and policy contexts in ways that earlier tools did not seem to support.

7 Discussion

Across the three study phases, a coherent picture emerges of how novice users make sense of air quality data. The novice survey revealed core misunderstandings about AQI concepts; expert interviews clarified which scientific ideas most needed to be communicated; and the iterative design cycles translated these needs into targeted design features. The public evaluation then showed how these choices played out in practice, with users moving beyond low-level struggles toward exploring patterns, contexts, and explanations. Together, these phases demonstrate how the study's multi-stage approach collectively shaped ContAQT's final design. In this section, we reflect on our research findings for **RQ1** from the perspective of oversimplification in science communication. We then outline design implications for interactive data visualization systems based on our findings for **RQ2**. Finally, we discuss the limitations of this work and suggest directions for future research.

7.1 The Pitfalls of Over-Simplification

Our analysis in Section 4 identified five LCD Tensions (Table 3). Across the five tensions, a common thread emerges: tools often sacrifice interpretability for simplicity. While simplification can make data look more approachable, in practice, it might strip away the very structures novices need to make sense of pollutant dynamics. This pattern highlights a broader challenge for science communication: simplified representations may risk obscuring novices' understanding of scientific concepts.

On the one hand, simplification is commonly used in science communication and was identified as valuable [17]. For example, the AQI's color-coded categories provide a rapid cue for daily health risks, making complex data actionable in everyday contexts [2]. The metaphor of the "ozone hole," widely used in the 1980s to describe the ozone layer declining in concentration, conveyed urgency and irreparable damage, shaping both public perception and scientific discourse [17]. Other index-based representations, such as the Saffir-Simpson Hurricane Scale [35], the UV Index [1], and the Heat Index [36], are also widely adopted because they condense complex, multidimensional scientific measurements into a single interpretable value for the public. These cases demonstrate that simplified indicators can be powerful when the goal is awareness, warning, and immediate decision-making.

Yet our study demonstrates the limitations of this approach. Discrete colors obscure AQI as a continuous scale; collapsing into a single value flattens the composite of pollutants; charts without context hinder purpose comprehension; health framings miss personal relevance; and generalized reports erase local and temporal detail. Similar limitations have been observed in other index systems. For instance, research on the Saffir-Simpson Hurricane Scale shows that collapsing multidimensional storm hazards into a single category can mislead people's judgments about risk and storm severity [39]. Together, these tensions show how simplification can reinforce misunderstanding rather than promote clarity. Prior science communication and visual communication echoes this danger [17, 24, 57], such as reinforcing misunderstanding rather than fostering comprehension [20] and distorting interpretation

because important details were missing [33]. In short, when simplification becomes the dominant design principle, it risks creating more misunderstanding than clarity.

Our design iterations suggest a more productive balance: provide simple, approachable entry points, but connect them to richer representations that expose the composite structure of AQI. In the ContAQT platform, this took the form of moving from heatmaps in AQI timeline to calendar-based breakdowns, from static scores to interactive pollutant views. These shifts show how design can still maintain accessibility without sacrificing interpretability.

Looking beyond air quality, this insight extends to science communication more broadly. Novices are not only information receivers but also citizens and decision-makers who draw on scientific data and concepts in their daily activities [10]. Our findings also highlight a more general challenge for index-based scientific representations beyond Air Quality Index. While these systems offer intuitive entry points, their reduced dimensionality can obscure the underlying mechanisms that novices actively seek to understand, revealing a gap between what index values are designed to summarize and the deeper questions that novices naturally raise when trying to make sense of and make decisions about scientific data. By framing pitfalls as LCD Tensions, we show how designers can anticipate when simplification will hinder learning and where richer representations are needed. For scientific communication, simplification should not be a design objective to achieve, but a strategy to help achieve the goal of more transparent, accessible, and engaging communication to empower novices.

7.2 Implications for Designing Data Visualization to Make Scientific Data More Accessible

Through our iterative design and evaluation process, we tried a set of strategies to create interactive data visualizations in informal learning spaces that can support the communication of scientific data. From these efforts, we derive several design implications to guide future efforts aimed at making complex scientific information accessible to novice users, with particular attention to informal learning contexts.

We derived the following design implications through the reflection of our iterative design process.

- **Balance familiar and novel visual encoding design.** The development of the AQI Core Chart, where the familiar bar chart was used to introduce the novel Rosa chart, proved to be effective in supporting novices' understanding. In this case, designers should balance familiar encodings with novel ones that reveal disciplinary perspectives, which is also mentioned in [26]. Familiar formats give novices confidence and a recognizable entry point, while novel representations expand their understanding. Introducing novel encodings gradually, after establishing familiarity, helps users build from what they know toward more complex ways of seeing data.
- **Layer interactions to distribute learning goals.** The access to the AQI Calendar view was redesigned to be embedded in AQI Timeline view and was proven to be effective in reducing novices' cognitive load. Complex data should be distributed across multiple levels of interaction rather than

concentrated in a single display. Prior work has discussed that a layered interaction structure could reduce cognitive overload, allow audiences to choose the depth of engagement, and help scaffold users' inquiry process [28]. Shallow overviews provide easy entry, while deeper layers invite motivated novices to pursue more detailed exploration at their own pace.

- **Make terminology explicit and accessible.** Disciplinary terminology should never be assumed as common knowledge [43]. When technical terms remain unexplained, novices face immediate barriers to interpretation. Embedding clear labels, legends, and pop-up information cards within the interface ensures that concepts are both visible and understandable, enabling audiences to build literacy without external support.

We derived the following design implications from public visitors' comments collected during the local science festival.

- **Surface potential misunderstandings.** Regarding AQI as taking the summary or average of all six pollutants instead of the maximum is a misconception, as mentioned in Section 6.3.1, and novices expressed surprise about the actual AQI computation approach. Previous work has shown that designers can leverage on museum visitors' existing misconceptions to spark inquiry and critical thinking [46]. In this way, potential misunderstandings should not be hidden but deliberately surfaced. Confusion of a concept can also act as a form of desirable difficulty [5, 19], where carefully calibrated challenges prompt novices to slow down, ask questions, and generate explanations. By making points of potential confusion visible and explorable, visualizations transform this uncertainty into motivation, encouraging novices to move from passive reception toward active sense-making and deeper engagement.
- **Leave space for unsolved questions to spark curiosity.** As mentioned in the new theme "Curiosity about data patterns" in Section 6.3.5. In visitors' comments, many of them ask questions about information beyond what is currently presented in the ContAQT platform. Rather than treating these as incomplete information that needed to be added to the system in a later version, we view them as opportunities to trigger visitors' curiosity. As mentioned in previous museum studies [16, 48], a brief experience in an informal learning scenario should not aim to answer everything about the field, but expose users to opportunities for questions to continue sense-making, seek out additional resources, and connect the exhibit to their own contexts. In this way, the informal learning experience supports not only immediate comprehension but also sustained curiosity beyond the exhibit.
- **Connect data to personal experiences to spark conversations.** Designers should recognize that people engage more deeply when data is personally relevant to them, as shown in the new theme Framing AQI in societal contexts in Section 6.3.5. As previous work [40] notes, audiences often look for connections to their own lives and communities. Embedding everyday contexts into scientific data makes abstract

values tangible, motivating users to ask questions, share perspectives, and initiate dialogue. By grounding information in personal experience, visualizations move beyond individual comprehension to foster collective meaning-making.

Although developed in the context of air quality, the research team believes these strategies hold broader relevance for communicating other forms of complex scientific data, such as climate change, genomics, or energy use. Scientific domains often face a similar Gulf of Expertise: experts seek to convey nuance, while novices encounter simplified summaries. Applying scaffolding strategies can make abstract processes approachable while also fostering curiosity and more in-depth inquiry through data.

7.3 Limitations and Future Work

This work has several limitations that point to directions for future research.

First, our participants in the novice survey and in-lab user studies were skewed toward higher-educated members of the public and were mostly urban and suburban participants, reflecting the demographics of the research institute's surrounding community, one of the primary locations where the kiosk is ultimately intended to be deployed. However, since even these participants showed difficulty interpreting air quality representations, it's likely that less educated citizens and people who live in more rural areas will have similar difficulties interpreting air quality data. In addition, while the public evaluation included users of a wide age range, from children to adults, our design goal in this phase was to support general novice visitors rather than to develop child-specific learning experiences. Accordingly, we analyzed comments from children and adults together. As we move toward wider deployment scenarios, we plan to include a more diverse sample in subsequent phases of this research and explicitly compare how younger audiences and adults from a wider range of educational and geographic backgrounds interpret and use AQI visualizations.

Second, our current system primarily presents AQI through a timeline, but offers limited insight into how air quality is interpreted within broader social and cultural contexts. As some users noted, AQI can also be understood in relation to events, seasonal activities, or public holidays, where its implications become more immediate and meaningful. Future versions of the kiosk could incorporate features that dynamically connect AQI patterns to these live contexts, enabling users to relate changes in air quality to what is happening around them. Beyond adding contextual information, future work should examine how such contextual cues shape users' sensemaking processes, such as how they prompt new questions, shift attention between data and lived experience, or transform abstract trends into personally relevant concerns. Developing more fine-grained analytic accounts of these interactions would deepen HCI insights into public data sensemaking while informing the design of systems that help people see not only "what the data says" but also "why it matters now." Embedding these connections could make air quality data more salient, personal, and actionable.

Third, while our evaluations captured users' experience, engagement, and questions they held towards our system, they did not formally assess learning gains or knowledge transfer, considering

the setup of the system in a pop-up event. Future work would focus on understanding whether interaction with the system leads to deeper conceptual understanding, increased environmental literacy, and long-term data literacy skills, which requires more structured evaluation.

Lastly, the current design is oriented toward individual exploration. Yet air quality is inherently a collective issue that benefits from shared reflection. We aim to create a data display in informal learning spaces that are also collaborative, supporting opportunities for social play, meaningful interaction, and collaboration [6]. It could expand the system's role from a personal tool to elicit joint sensemaking. Future work may investigate how to leverage the large screen size for multi-touch, multi-user interaction and how collaborative displays in a shared public space can encourage shared knowledge and extrinsic motivation to alter behavior in response to presented data.

8 Conclusion

In conclusion, this work introduced ContAQT, an interactive kiosk designed through a learner-centered design approach to make multi-pollutant air quality data accessible in informal learning spaces. By identifying design tensions in existing communication tools, iteratively prototyping data visualization features, and evaluating the system during a public science festival, we showed how visualization designs can address pitfalls in existing tools for communicating air quality data, foster inquiry, and create opportunities for sense-making around environmental and data issues. Our findings highlight the double-edged role of simplification in science communication and provide design implications for public-facing data visualizations to make scientific data accessible. Looking forward, we see opportunities to extend this work toward measuring learning growth and exploring how collaborative public displays can spark learning discourse around environmental challenges.

Acknowledgments

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References

- [1] Environmental Protection Agency. 2015. The UV Index. <https://www.epa.gov/sunsafety/uv-index-1>. Accessed: 2025-11-25.
- [2] Environmental Protection Agency. 2018. Criteria Air Pollutants. <https://www.epa.gov/criteria-air-pollutants>. Accessed: 2025-11-25.
- [3] Environmental Protection Agency. 2018. *Technical Assistance Document for the Reporting of Daily Air Quality – the Air Quality Index (AQI)*. Environmental Protection Agency. 5–7 p. pages.
- [4] Environmental Protection Agency. 2020. Our Nation's Air: Status and Trends Through 2019. Accessed: 2025-11-25.
- [5] Elizabeth L Bjork, Robert A Bjork, et al. 2011. Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning. *Psychology and the real world: Essays illustrating fundamental contributions to society* 2, 59–68 (2011), 56–64.
- [6] Florian Block, Michael S Horn, Brenda Caldwell Phillips, Judy Diamond, E Margaret Evans, and Chia Shen. 2012. The deepree exhibit: Visualizing the tree of life to facilitate informal learning. *IEEE Transactions on Visualization and Computer Graphics* 18, 12 (2012), 2789–2798.
- [7] Katy Börner, Adam Maltese, Russell Nelson Balliet, and Joe Heimlich. 2016. Investigating aspects of data visualization literacy using 20 information visualizations and 273 science museum visitors. *Information Visualization* 15, 3 (2016), 198–213.
- [8] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [9] Kenneth Broad, Anthony Leiserowitz, Jessica Weinkle, and Marissa Steketee. 2007. Misinterpretations of the “cone of uncertainty” in Florida during the 2004 hurricane season. *Bulletin of the American Meteorological Society* 88, 5 (2007), 651–668.
- [10] Terry W Burns, D John O'Connor, and Susan M Stockmayer. 2003. Science communication: a contemporary definition. *Public understanding of science* 12, 2 (2003), 183–202.
- [11] National Research Council. 2009. *Learning Science in Informal Environments: People, Places, and Pursuits*. The National Academies Press, Washington, DC. doi:10.17226/12190
- [12] Keshav Dasu, Kwan-Liu Ma, Joyce Ma, and Jennifer Frazier. 2020. Sea of genes: A reflection on visualising metagenomic data for museums. *IEEE Transactions on Visualization and Computer Graphics* 27, 2 (2020), 935–945.
- [13] Pryce Davis, Michael Horn, Florian Block, Brenda Phillips, E. Margaret Evans, Judy Diamond, and Chia Shen. 2015. “whoa! we’re going deep in the trees!”: Patterns of collaboration around an interactive information visualization exhibit - International Journal of Computer-supported collaborative learning. <https://link.springer.com/article/10.1007/s11412-015-9209-z>
- [14] Magali Delmas and Aanchal Kohli. 2020. Can Apps Make Air Pollution Visible? Learning About Health Impacts Through Engagement with Air Quality Information. *Journal of Business Ethics* 161 (01 2020). doi:10.1007/s10551-019-04215-7
- [15] Sara Doan. 2021. Misrepresenting COVID-19: Lying with charts during the second golden age of data design. *Journal of Business and Technical Communication* 35, 1 (2021), 73–79.
- [16] John H Falk and Lynn D Dierking. 2016. *The museum experience revisited*. Routledge.
- [17] Reiner Grundmann and Jean-Pierre Cavaillé. 2000. Simplicity in Science and its Publics. *Science as Culture* 9, 3 (2000), 353–389.
- [18] Chien-Hsin Hsueh, Jacqueline Chu, Kwan-Liu Ma, Joyce Ma, and Jennifer Frazier. 2016. Fostering comparisons: Designing an interactive exhibit that visualizes marine animal behaviors. In *2016 IEEE Pacific Visualization Symposium (PacificVis)*. 259–263. doi:10.1109/PACIFICVIS.2016.7465282
- [19] Jessica Hullman, Eytan Adar, and Priti Shah. 2011. Benefitting infovis with visual difficulties. *IEEE Transactions on Visualization and Computer Graphics* 17, 12 (2011), 2213–2222.
- [20] Dan Kahan. 2012. Why we are poles apart on climate change. *Nature* 488, 7411 (2012), 255–255.
- [21] Sunyoung Kim and Muyang Li. 2020. Awareness, Understanding, and Action: A Conceptual Framework of User Experiences and Expectations about Indoor Air Quality Visualizations. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3313831.3376521
- [22] Sunyoung Kim, Muyang Li, Jennifer Senick, and Gediminas Mainelis. 2020. Designing to Engage Children in Monitoring Indoor Air Quality: A Participatory Approach. In *Proceedings of the Interaction Design and Children Conference* (London, United Kingdom) (IDC '20). Association for Computing Machinery, New York, NY, USA, 323–334. doi:10.1145/3392063.3394395
- [23] Sunyoung Kim and Eric Paulos. 2010. InAir: sharing indoor air quality measurements and visualizations. (01 2010), 1861–1870. doi:10.1145/1753326.1753605
- [24] Xingyu Lan and Yu Liu. 2024. “I Came Across a Junk”: Understanding Design Flaws of Data Visualization from the Public's Perspective. *IEEE Transactions on Visualization and Computer Graphics* (2024).
- [25] Jongwook Lee, Gary Burnett, Micah Vandegrift, Jung Hoon Baeg, and Richard Morris. 2015. Availability and accessibility in an open access institutional repository: a case study. *Information Research: An International Electronic Journal* 20, 1 (2015), n1.
- [26] Hsin-I Liao, Su-Ling Yeh, and Shinsuke Shimojo. 2011. Novelty vs. familiarity principles in preference decisions: task-context of past experience matters. *Frontiers in psychology* 2 (2011), 43.
- [27] Andrés Lucero. 2015. Using affinity diagrams to evaluate interactive prototypes. In *IFIP conference on human-computer interaction*. Springer, 231–248.
- [28] J. Ma, I Liao, K. Ma, and J. Frazier. 2012. Living Liquid: Design and Evaluation of an Exploratory Visualization Tool for Museum Visitors. *IEEE Transactions on Visualization and Computer Graphics* 18, 12 (2012), 2799–2808. doi:10.1109/TVCG.2012.244
- [29] Ellen B. Mandinach and Edith S. Gummer. 2013. A Systemic View of Implementing Data Literacy in Educator Preparation. *Educational Researcher* 42, 1 (2013), 30–37. arXiv:<https://doi.org/10.3102/0013189X12459803> doi:10.3102/0013189X12459803
- [30] Eric M Meyers, Ingrid Erickson, and Ruth V Small. 2013. Digital literacy and informal learning environments: an introduction. *Learning, media and technology* 38, 4 (2013), 355–367.
- [31] David Mintz. 2009. Technical assistance document for the reporting of daily air quality—the air quality index (AQI). *Tech. Research Triangle Park, US Environmental Protection Agency* (2009).
- [32] Jimmy Moore, Pascal Goffin, Miriah Meyer, Philip Lundrigan, Neal Patwari, Katherine Sward, and Jason Wiese. 2018. Managing In-Home Environments through Sensing, Annotating, and Visualizing Air Quality Data. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 2, 3, Article 128 (Sept. 2018), 28 pages. doi:10.1145/3264938

- [33] Vinh T Nguyen, Kwanghee Jung, and Vibhuti Gupta. 2021. Examining data visualization pitfalls in scientific publications. *Visual Computing for Industry, Biomedicine, and Art* 4, 1 (2021), 27.
- [34] Donald A. Norman. 1993. *Things that make us smart: defending human attributes in the age of the machine*. Addison-Wesley Longman Publishing Co., Inc., USA.
- [35] National Oceanic and Atmospheric Administration. 2012. The Saffir-Simpson Hurricane Wind Scale. <https://www.epa.gov/sunsafety/uv-index-1>. Accessed: 2025-11-25.
- [36] National Oceanic and Atmospheric Administration. 2024. What is the heat index? <https://www.weather.gov/ama/heatindex>. Accessed: 2025-11-25.
- [37] World Health Organization. 2016. *Ambient air pollution: a global assessment of exposure and burden of disease*. World Health Organization. 121 p. pages.
- [38] Seymour Papert. 1993. *The children's machine: Rethinking school in the age of the computer*. Basic Books, Inc.
- [39] Leilani D Paxton, Jennifer Collins, and Laura Myers. 2024. Reconsidering the Saffir-Simpson Scale: A Qualitative Investigation of Public Understanding and Alternative Frameworks. In *Advances in Hurricane Risk in a Changing Climate*. Springer, 241–279.
- [40] Evan M Peck, Sofia E Ayuso, and Omar El-Etr. 2019. Data is personal: Attitudes and perceptions of data visualization in rural pennsylvania. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–12.
- [41] Lars Perlmutter, David Stieb, and Kevin Cromar. 2017. Accuracy of quantification of risk using a single-pollutant Air Quality Index. *Journal of Exposure Science and Environmental Epidemiology* 27, 1 (1 Jan 2017), 24–32. doi:10.1038/jes.2015.43
- [42] Jean Piaget. 2013. *The construction of reality in the child*. Routledge.
- [43] Zachary Pousman, John Stasko, and Michael Mateas. 2007. Casual information visualization: Depictions of data in everyday life. *IEEE transactions on visualization and computer graphics* 13, 6 (2007), 1145–1152.
- [44] Qualtrics. 2024. Qualtrics: Online Survey Software. <https://www.qualtrics.com>. Accessed: 2025-12-01.
- [45] Chris Quintana, Joseph Krajcik, and Elliot Soloway. 2013. Exploring a structured definition for learner-centered design. In *International Conference of the Learning Sciences*. Psychology Press, 256–263.
- [46] Stella Quinto Lima, Gabriela Buraglia, Wong Kam-Kwai, and Jessica Roberts. 2025. Data Bias Recognition in Museum Settings: Framework Development and Contributing Factors. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 229, 15 pages. doi:10.1145/3706598.3714092
- [47] Jessica Roberts and Leilah Lyons. 2017. Scoring Qualitative Informal Learning Dialogue: The SQuILD Method for Measuring Museum Learning Talk. Philadelphia, PA: International Society of the Learning Sciences.
- [48] Jessica Roberts and Leilah Lyons. 2017. The value of learning talk: applying a novel dialogue scoring method to inform interaction design in an open-ended, embodied museum exhibit. *International Journal of Computer-Supported Collaborative Learning* 12, 4 (2017), 343–376.
- [49] Nina Simon. 2010. *The participatory museum*. Museum 2.0.
- [50] Elliot Soloway, Mark Guzdial, and Kenneth E Hay. 1994. Learner-centered design: The challenge for HCI in the 21st century. *interactions* 1, 2 (1994), 36–48.
- [51] Elliot Soloway, Shari L Jackson, Jonathan Klein, Chris Quintana, James Reed, Jeff Spitulnik, Steven J Stratford, Scott Studer, Jim Eng, and Nancy Scala. 1996. Learning theory in practice: Case studies of learner-centered design. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. 189–196.
- [52] Yolande AA Strengers. 2011. Designing eco-feedback systems for everyday life. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 2135–2144.
- [53] Marcus Winter. 2018. Visitor perspectives on commenting in museums. *Museum Management and Curatorship* 33, 5 (2018), 484–505.
- [54] Annika Wolff, Daniel Gooch, Jose J Caverio Montaner, Umar Rashid, and Gerd Kortuem. 2016. Creating an understanding of data literacy for a data-driven society. *The Journal of Community Informatics* 12, 3 (2016).
- [55] Jung-Hun Woo, Sang Lim, Karpjoo Jeong, HyungSeok Kim, Jae-Jin Kim, JongHyun Lee, Junghee Kim, Taehoon Lee, Minh Le, Rina Ryoo, Suhyang Kim, Hansoo Kim, and Jee-In Kim. 2008. AirScope: A Micro-Scale Urban Air Quality Management System. 378–379. doi:10.1109/eScience.2008.125
- [56] Keke Wu, Emma Petersen, Tahmina Ahmad, David Burlinson, Shea Tanis, and Danielle Albers Szafir. 2021. Understanding data accessibility for people with intellectual and developmental disabilities. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–16.
- [57] C. Xiong, L. Van Weelden, and S. Franconeri. 2020. The Curse of Knowledge in Visual Data Communication. *IEEE Transactions on Visualization and Computer Graphics* 26, 10 (2020), 3051–3062. doi:10.1109/TVCG.2019.2917689
- [58] Huan Xu, Wei Zeng, Bing Guo, Philip K. Hopke, Xue Qiao, Hyunok Choi, Bin Luo, Wei Zhang, and Xing Zhao. 2020. Improved risk communications with a Bayesian multipollutant Air Quality Health Index. *Science of The Total Environment* 722 (2020), 137892. doi:10.1016/j.scitotenv.2020.137892